Agent_1

https://gemini.google.com/app/b174016cb3aa2181

User prompt: you are part of a big mission and your task is given in the md file

Response: Spatial Domain & Boundary Conditions

- **Spatial Domain:** The system is defined over $x \in (-\infty, \infty)$ and $y \in [0, \infty)$.
- **Ground Boundary Condition ($y = 0$):** Because the potential energy is infinite for $y < 0$, the wave function must vanish at $y = 0$ to maintain continuity. This imposes the condition $\psi(x, 0) = 0 \implies Y(0) = 0$.
- **Asymptotic Boundary Condition ($y \to \infty$):** As $y \to \infty$, $V(x, y) = mgy \to \infty$. Physical normalizability requires that $\lim_{y \to \infty} \psi(x, y) = 0 \implies \lim_{y \to \infty} Y(y) = 0$.
- **Horizontal Domain ($x \in (-\infty, \infty)$):** Since no potential barrier exists in the horizontal direction ($V_x(x) = 0$), $X(x)$ represents free translational motion along the x-axis and remains bounded for all $x \in (-\infty, \infty)$.

Classical Energy Decomposition

- The initial speed is u at launch angle θ under uniform gravitational acceleration $g = 10\ \mathrm{m/s^2}$.
- The initial velocity components are $v_x = u\cos\theta$ and $v_y = u\sin\theta$.
- **Horizontal Kinetic Energy Component:**

$$E_x = \frac{1}{2}m(u\cos\theta)^2$$

- **Vertical Kinetic Energy Component:**

$$E_y = \frac{1}{2}m(u\sin\theta)^2$$

- The total system energy is $E = E_x + E_y = \frac{1}{2}mu^2$.

2D Hamiltonian Operator

- The 2D Hamiltonian operator $\hat{H}$ for a particle of mass m in uniform gravity $g = 10\ \mathrm{m/s^2}$ is:

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(x, y) = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + V(x, y)$$

- The potential $V(x, y)$ is defined piecewise:

$$V(x, y) = \begin{cases} mgy, & \text{for } y \geq 0 \\ \infty, & \text{for } y < 0 \end{cases}$$

- For $y \geq 0$, the Time-Independent Schrödinger Equation $\hat{H}\psi(x, y) = E\psi(x, y)$ takes the form:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}\right)+mgy\psi(x,y)=E\psi(x,y)$$

Separation of Variables

Substituting the separable ansatz $\psi(x,y)=X(x)Y(y)$ into the 2D Schrödinger equation yields:

$$-\frac{\hbar^2}{2m}\left(Y(y)\frac{d^2X(x)}{dx^2}+X(x)\frac{d^2Y(y)}{dy^2}\right)+mgyX(x)Y(y)=EX(x)Y(y)$$

Dividing the entire equation by $X(x)Y(y)$:

$$-\frac{\hbar^2}{2m}\frac{1}{X(x)}\frac{d^2X(x)}{dx^2}-\frac{\hbar^2}{2m}\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2}+mgy=E$$

Rearranging terms separates x-dependent components from y-dependent components:

$$-\frac{\hbar^2}{2m}\frac{1}{X(x)}\frac{d^2X(x)}{dx^2}=E-\left(-\frac{\hbar^2}{2m}\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2}+mgy\right)$$

Since the left side depends exclusively on x and the right side depends exclusively on y, both sides must equal a constant separation parameter E_x. Setting $E_y=E-E_x$, we decouple the partial differential equation into two independent 1D Ordinary Differential Equations (ODEs):

1. **Horizontal ODE ($X(x)$):**

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2}=E_xX(x)\implies\frac{d^2X(x)}{dx^2}+k_x^2X(x)=0$$

   where $k_x=\frac{2mE_x}{\hbar}=\frac{mu\cos\theta}{\hbar}$.

2. **Vertical ODE ($Y(y)$ for $y\geq0$):**

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2}+mgyY(y)=E_yY(y)\implies\frac{d^2Y(y)}{dy^2}+\frac{2m}{\hbar^2}\left(E_y-mgy\right)Y(y)=0$$

   where $E_y=\frac{1}{2}m(u\sin\theta)^2$.
